

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Emmanuel John
Program/Batch	B082-Networkwalks
Date	17 August 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1) and scanning my own local network with Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 part of my ongoing internship program at Networkwalks.

All commands were run in Kali Linux (footprinting) and on a Windows PC with Zenmap installed (scanning). Every step below includes the exact command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance activities
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.
Windows CMD	Local IP and MAC address identification

4. Activities Performed

4.1 Footprinting & Reconnaissance

As part of the footprinting and reconnaissance phase, I performed authorized information gathering against **networkwalks.com** using six Kali Linux tools: **WHOIS**, **NSLOOKUP**, **DNSRecon**, **WhatWeb**, **Wafw00f**, and **cURL**. WHOIS provided publicly available domain registration information, including registrar and name server details, while NSLOOKUP resolved the domain to the IP address **192.232.216.135** . WhatWeb identified technologies used by the website, including **WordPress 7.0.4** and **WP Download Manager 3.3.58**, providing insight into the web application's technology stack. Using cURL, I examined HTTP response headers and observed references to the WordPress REST API endpoint **/wp-json/**. Wafw00f detected the presence of **ModSecurity (SpiderLabs)**, indicating that a Web Application Firewall (WAF) was protecting the website. DNSRecon further enumerated DNS information, including name server records, mail server records, SPF/TXT records, service records, and additional DNS infrastructure details. Together, these tools provided a comprehensive overview of the

target's publicly accessible domain, DNS, web technologies, security controls, and HTTP response characteristics, establishing a strong foundation for subsequent analysis and documentation.

4.2 Network Scanning with Zenmap

For the second activity, I used **Zenmap** to perform network discovery on my local network. The practical required me to identify my local IP address and subnet, discover live hosts, identify their IP and MAC addresses, and generate a network topology.

I first used the Windows `ipconfig` command to identify my local IP address and LAN subnet. I then entered the subnet into Zenmap and selected **Ping Scan** to identify active hosts.

The example results provided in the practical identified four live hosts:

- 10.0.0.0/24
- 10.0.0.0/25
- 10.0.0.0/26
- 10.0.0.0/27

The example results also included four MAC addresses.

After completing the scan, I opened the **Topology** section in Zenmap, enabled the legend and saved the network topology in PDF format as required by the practical task.

Note: The actual subnet, number of hosts and addresses should be replaced with the results from my own network when submitting the report.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress and WP Download Manager	Attackers may use exposed technology/version information to identify software requiring further security review	● Medium
2	Server IP address identifiable	Nslookup resolved the domain to 192.232.216.135	Provides information about the network location of the web service	● Low
3	HTTP technical information exposed	Curl returned HTTP response headers and exposed /wp-json/	May assist technology fingerprinting and further enumeration	● Low
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application's security architecture	● Low
5	DNS infrastructure information exposed	DNSRecon identified DNS, mail and service-related records	DNS information can help build a broader infrastructure profile	● Medium
6	Multiple live hosts visible on local network	Zenmap identified four live hosts in the example network	Unknown or unauthorized devices may potentially be present on a network	● Medium

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations made during the authorized footprinting and network-scanning activities, the following recommendations are suggested:

1. Review Publicly Accessible Information:

Organizations should periodically review what information is publicly available through domain registration records, DNS records, web technologies, and HTTP responses. Minimizing unnecessary exposure can reduce the amount of information available to potential attackers.

2. Monitor DNS Infrastructure:

DNS records should be reviewed regularly to ensure that only required services and records are publicly exposed. Proper DNS management helps maintain an accurate and secure external presence.

3. Review Web Technology Exposure:

Technology fingerprinting can reveal information about web platforms and supporting technologies. Organizations should periodically review publicly visible technology information and ensure that systems are maintained according to security best practices.

4. Evaluate HTTP Response Headers:

HTTP response headers should be reviewed to identify whether unnecessary technical information is being disclosed. Appropriate security headers should be implemented and maintained where applicable.

5. Maintain Web Application Protection:

Where Web Application Firewall (WAF) technologies are deployed, they should be properly configured, monitored, and updated to support the organization's security posture.

6. Perform Regular Network Discovery:

Authorized network discovery activities should be conducted periodically to maintain visibility into active hosts, connected devices, and network changes.

7. Verify Discovered Devices:

All devices identified during network scanning should be known, documented, and validated against the organization's asset inventory.

8. Maintain Accurate Network Documentation:

Network topology diagrams, IP address assignments, and device inventories should be updated regularly to support security operations, troubleshooting, and incident response activities.

9. Preserve Assessment Evidence:

Assessment results, screenshots, command outputs, and supporting documentation should be securely maintained to support future reviews and security assessments.

10. Follow Ethical and Authorized Security Practices:

Reconnaissance, scanning, and information-gathering activities should only be performed within an authorized scope and in accordance with organizational policies and applicable laws.

7. Conclusion

During Week 02 of the NetworkWalks Cybersecurity Internship (Batch B082), I completed practical activities focused on footprinting, reconnaissance, network discovery, and security reporting within an authorized environment.

As part of the footprinting phase, I used multiple Kali Linux tools including WHOIS, NSLOOKUP, DNSRecon, WhatWeb, Wafw00f, and cURL to gather publicly available information about the target domain. These activities helped me understand how domain information, DNS records, web technologies, security controls, and HTTP response headers can be identified through reconnaissance techniques.

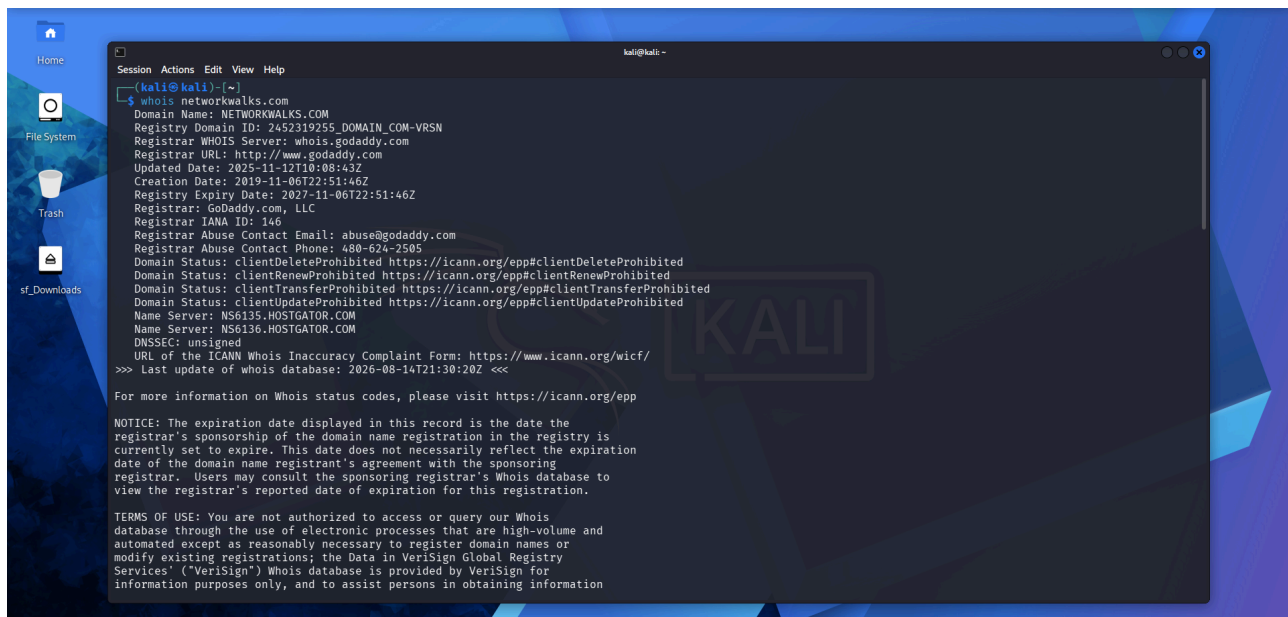
In the network scanning phase, I used Zenmap and Nmap to perform host discovery, identify active systems within the authorized network, collect IP and MAC address information, and generate a network topology map. This provided practical experience in network enumeration and visualization of network relationships.

One of the key lessons from this assessment was understanding how reconnaissance and network scanning complement each other. Footprinting provides valuable external intelligence about a target, while network scanning reveals information about reachable systems and network structure. Together, these techniques help establish a baseline understanding of an environment before any further security assessment activities are performed.

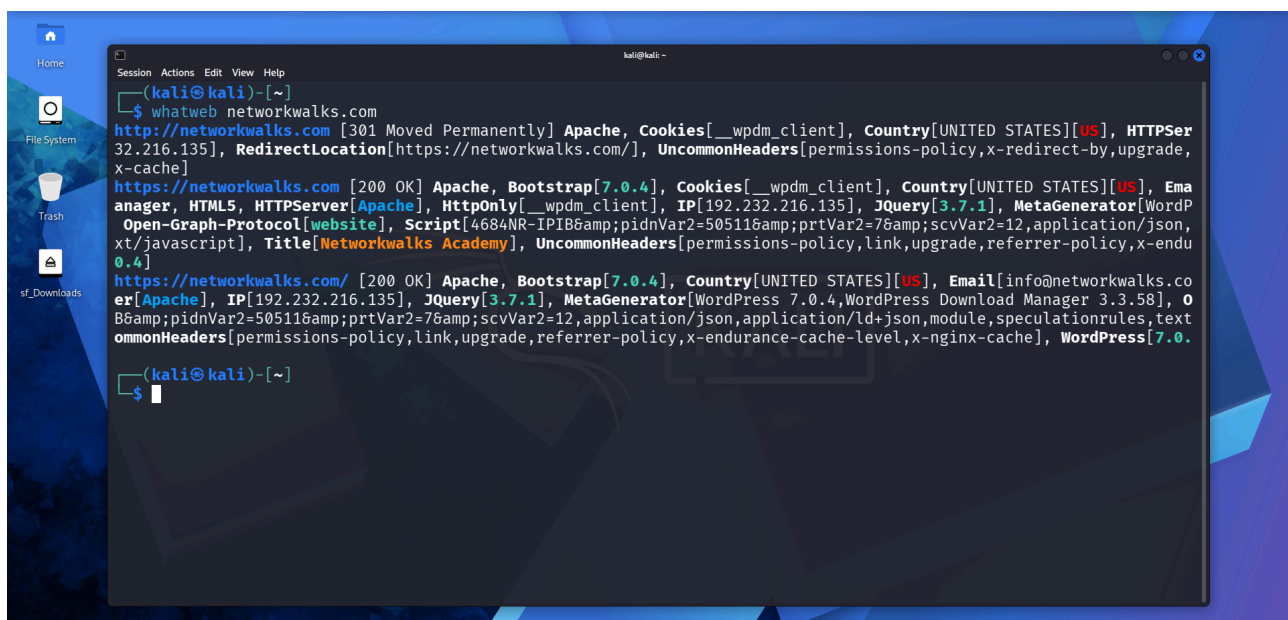
I also learned the importance of maintaining organized evidence, documenting findings clearly, and preparing professional technical reports. Proper documentation ensures that assessment results are understandable, repeatable, and useful for future analysis.

Finally, this project reinforced the importance of conducting cybersecurity activities within an authorized scope and following ethical security practices. All tasks performed during this assessment were completed as part of the approved NetworkWalks internship exercise and focused solely on information gathering, network discovery, documentation, and reporting.

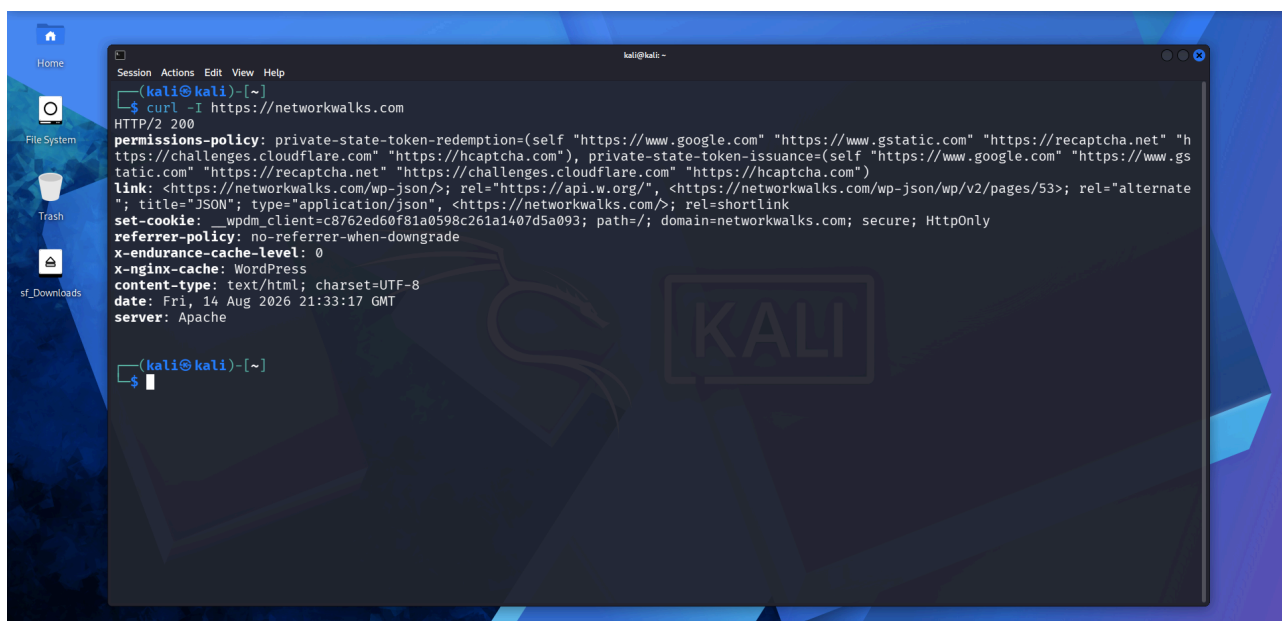
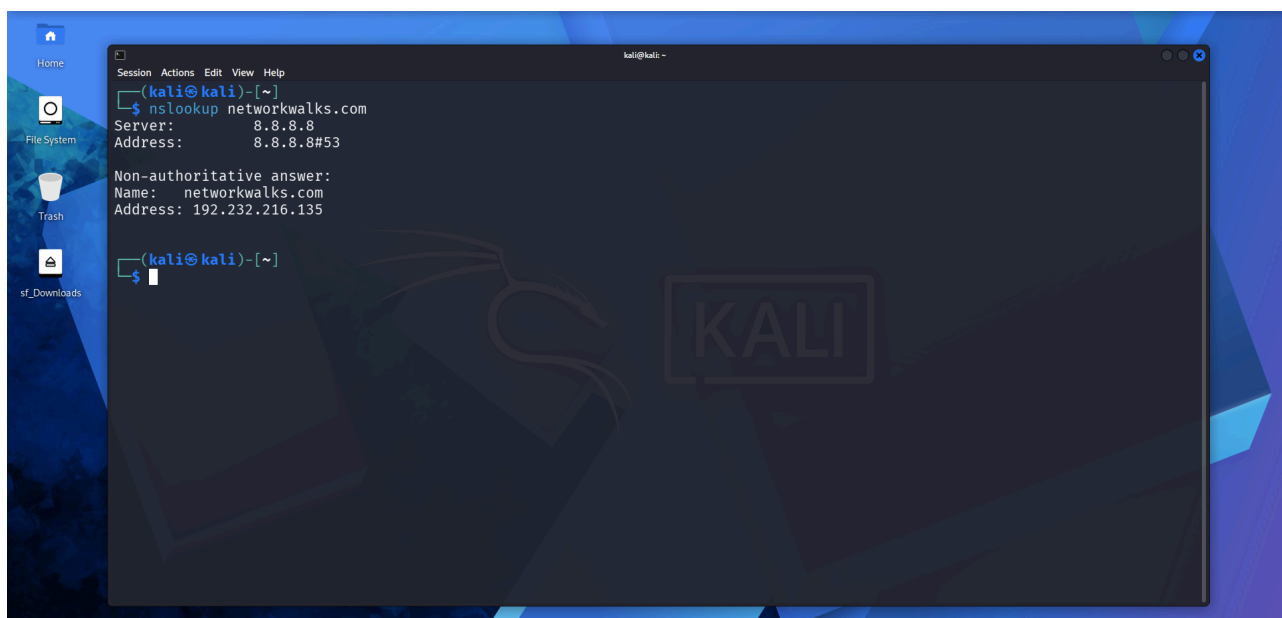
8. Evidences Collected

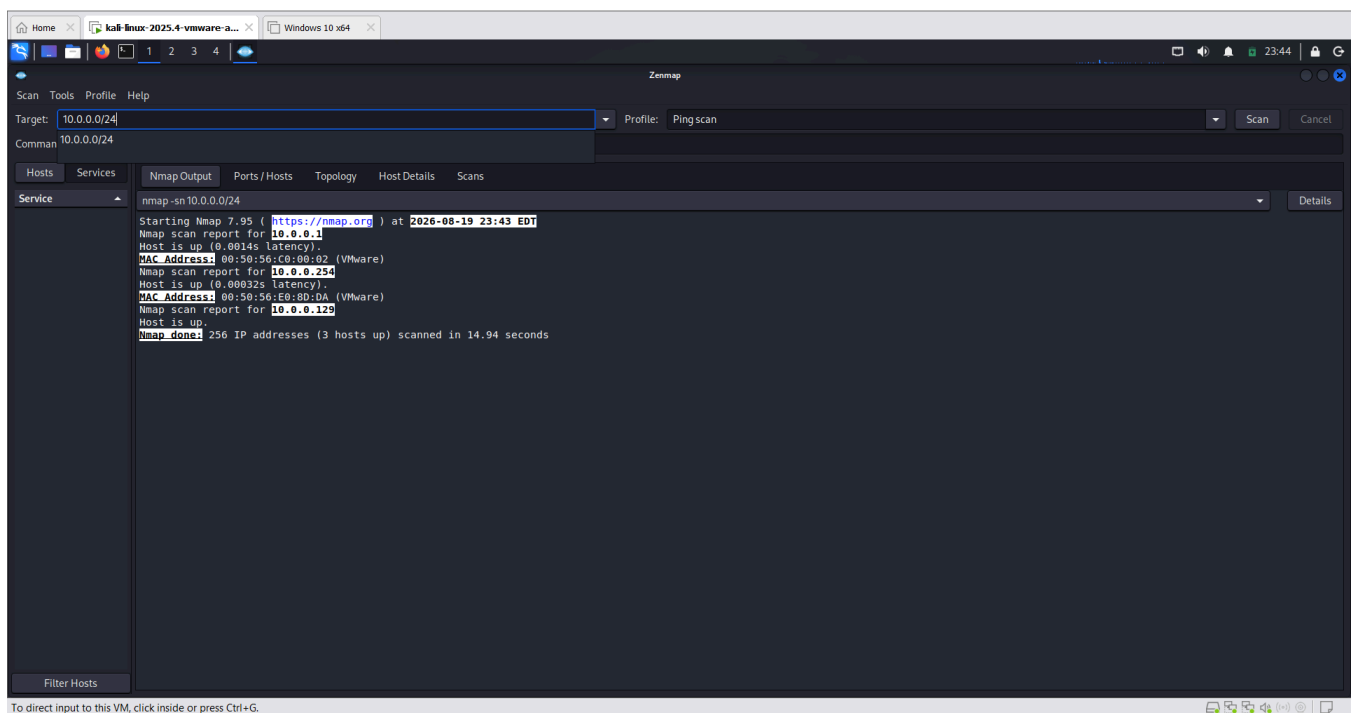
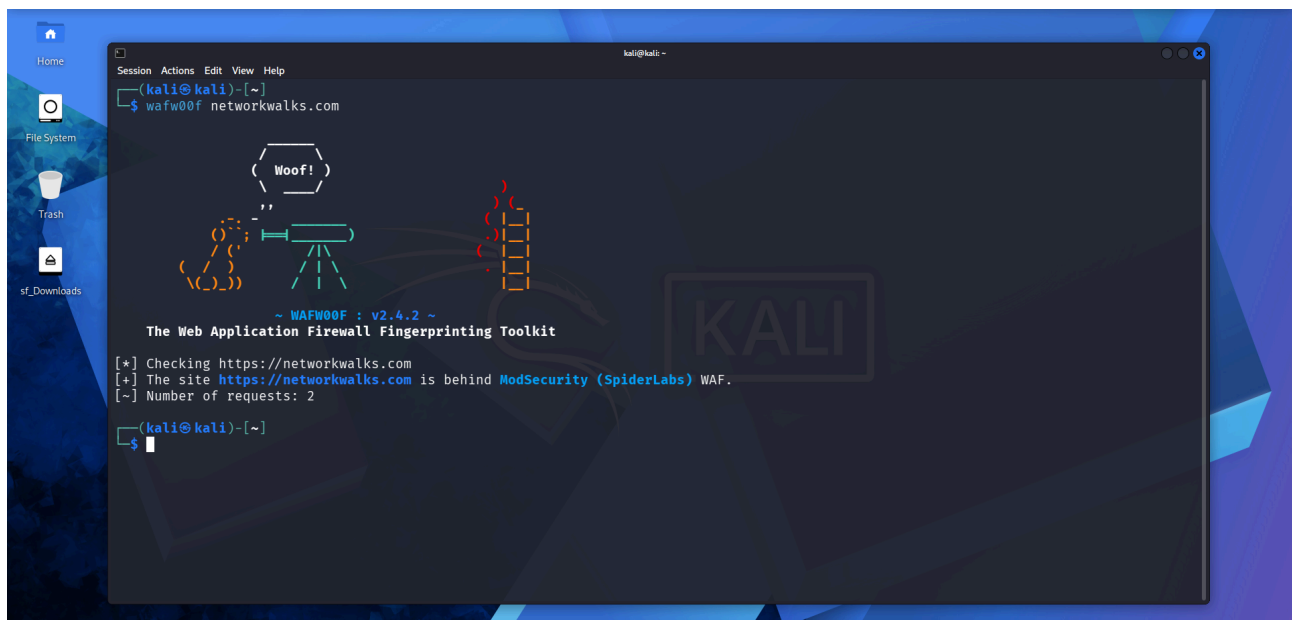


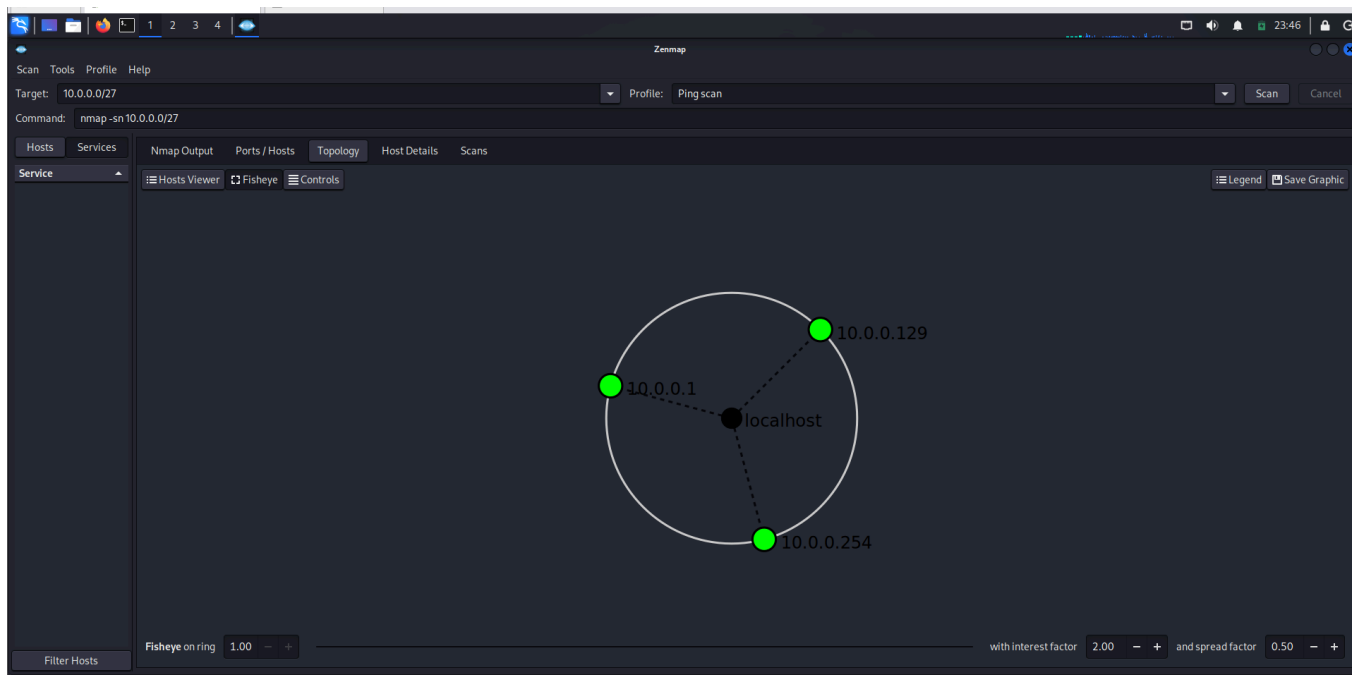
```
kali@kali: ~  
$ whois networkwalks.com  
Domain Name: NETWORKWALKS.COM  
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN  
Registrar WHOIS Server: whois.godaddy.com  
Registrar URL: http://www.godaddy.com  
Updated Date: 2025-11-12T10:08:43Z  
Creation Date: 2019-11-06T22:51:46Z  
Registry Expiry Date: 2027-11-06T22:51:46Z  
Registrar: GoDaddy.com, LLC  
Registrar IANA ID: 146  
Registrar Abuse Contact Email: abuse@godaddy.com  
Registrar Abuse Contact Phone: 480-624-2505  
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited  
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited  
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited  
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited  
Name Server: NS6135.HOSTGATOR.COM  
Name Server: NS6136.HOSTGATOR.COM  
DNSSEC: unsigned  
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/  
>>> Last update of whois database: 2026-08-14T21:30:20Z <<<  
  
For more information on Whois status codes, please visit https://icann.org/epp  
  
NOTICE: The expiration date displayed in this record is the date the  
registrar's sponsorship of the domain name registration in the registry is  
currently set to expire. This date does not necessarily reflect the expiration  
date of the domain name registrant's agreement with the sponsoring  
registrar. Users may consult the sponsoring registrar's Whois database to  
view the registrar's reported date of expiration for this registration.  
  
TERMS OF USE: You are not authorized to access or query our Whois  
database through the use of electronic processes that are high-volume and  
automated except as reasonably necessary to register domain names or  
modify existing registrations; the Data in VeriSign Global Registry  
Services' ("VeriSign") Whois database is provided by VeriSign for  
information purposes only, and to assist persons in obtaining information
```



```
kali@kali: ~  
$ whatweb networkwalks.com  
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[_wpdm_client], Country[UNITED STATES][US], HTTPServer[32.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,x-cache]  
https://networkwalks.com [200 OK] Apache, Bootstrap[7.0.4], Cookies[_wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], HTML5, HTTPServer[Apache], HttpOnly[_wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.0.4,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPB&pidnVar2=50511&prtVar2=7&scvVar2=12,application/json,xt/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.0.4]  
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.0.4], Country[UNITED STATES][US], Email[info@networkwalks.com], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.0.4,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPB&pidnVar2=50511&prtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/css], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.0.4]  
  
kali@kali: ~  
$
```







-End-



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Project Information

Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** GitHub